

OmicsTrust VANISH PC11 Report

A de novo transcriptomic vasopressin-response candidate axis in septic shock

Dataset: VANISH / E-MTAB-7581 whole-blood transcriptomics

Clinical context: Randomized septic shock trial comparing vasopressin vs norepinephrine, with factorial hydrocortisone/placebo exposure.

Analysis question: Is there a transcriptomic axis, beyond published SRS status, that modifies the association between vasopressin vs noradrenaline and 28-day mortality?

Metric	Result
Patients analyzed	116
Expression features	28,220
Screened PCs	25
Top candidate	PC11
Primary interaction OR per 1 SD PC11	0.169 (95% CI 0.051-0.563)
Primary LRT p / FDR	0.00126 / 0.0315
Permutation p	0.0040
Bootstrap direction stability	99.9% negative beta
Metadata explanation of PC11	R2 = 0.018; model p = 0.997

Bottom-line interpretation: PC11 is a research-stage candidate treatment-response axis. In this dataset, PC11-high patients had lower observed 28-day mortality with vasopressin than with noradrenaline, while PC11-low patients showed the opposite pattern. The interaction survives permutation testing, bootstrap stability checks, SRS-interaction challenges, and major-comorbidity adjustment. This is not a clinical recommendation and requires external validation.

Prepared for scientific/clinical discussion. Research use only. Not for bedside treatment selection.

1. Executive Summary

OmicsTrust performed a de novo treatment-response screen in the VANISH septic shock transcriptomic dataset. The screen evaluated whether any transcriptomic principal component modified the association between vasopressin versus noradrenaline and 28-day mortality, while adjusting for steroid exposure, SRS endotype, age, APACHE II score, and sex.

The strongest candidate was **PC11**. The vasopressin x PC11 interaction had OR 0.169 per 1 SD PC11, Wald $p=0.00377$, likelihood-ratio $p=0.00126$, and Benjamini-Hochberg FDR=0.0315 across 25 screened PCs. A permutation test of the top interaction produced $p=0.0040$, and bootstrap resampling showed a negative interaction coefficient in 99.9% of successful bootstrap samples.

The observed subgroup pattern was clinically interpretable: PC11-high patients had lower mortality with vasopressin than with noradrenaline, whereas PC11-low patients had higher mortality with vasopressin than with noradrenaline. Metadata explanation for PC11 was low ($R^2=0.018$, model $p=0.997$), and the interaction remained stable after adding known SRS-related treatment interactions and major comorbidities.

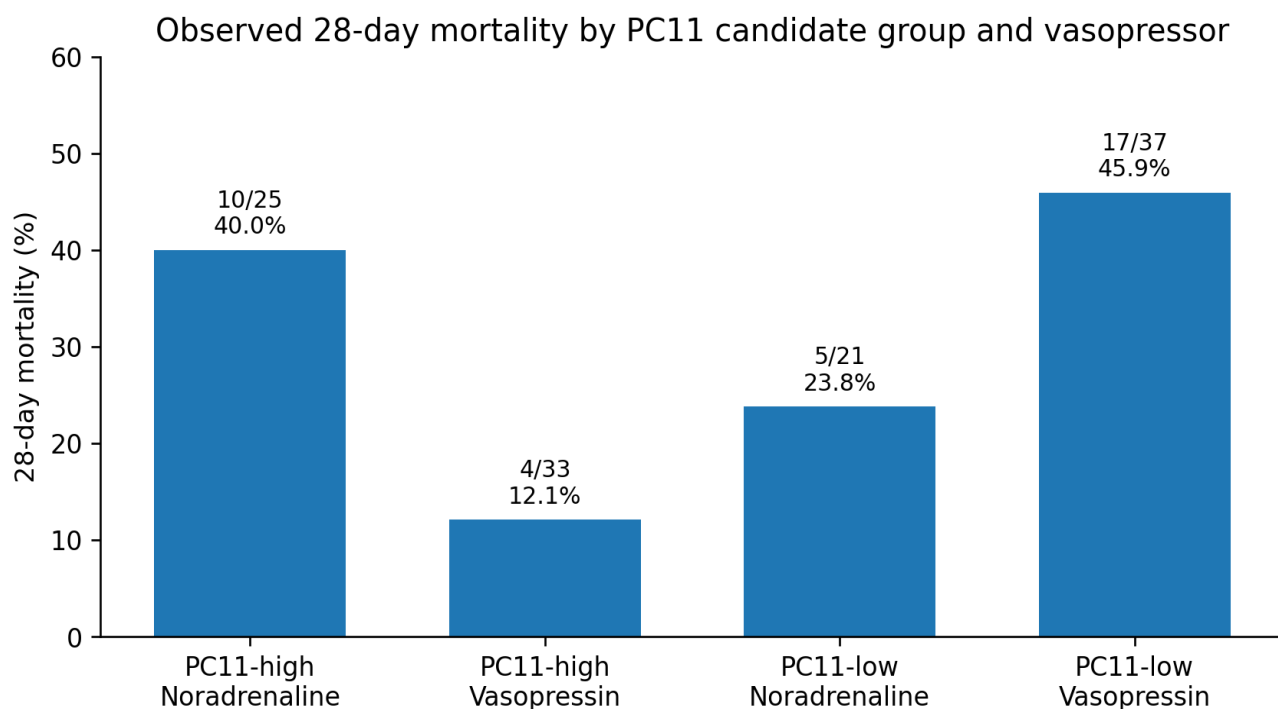


Figure 1. Observed 28-day mortality rates by PC11 candidate group and vasopressor assignment. Percentages are descriptive and should not be interpreted as causal estimates.

2. Dataset and Methods

Dataset. The analysis used the VANISH trial transcriptomic dataset E-MTAB-7581. VANISH was a double-blind randomized septic shock trial in which patients were randomized to vasopressin or norepinephrine, followed by hydrocortisone or placebo in the factorial design. VANISH transcriptomic data have also been used to assign Sepsis Response Signature (SRS) endotypes and evaluate differential treatment responses.

Input data. The OmicsTrust file used 116 patients and 28,220 expression features. The discovery screen selected the 5,000 most variable features and screened 25 principal components.

Primary model. For each PC, OmicsTrust fit a logistic interaction model for 28-day mortality:

death ~ vasopressor * PC + steroid + SRS + age + APACHE II + sex

Candidate selection. PCs were ranked by likelihood-ratio test for the vasopressor x PC interaction. Multiple testing was handled with Benjamini-Hochberg FDR across 25 screened PCs.

Robustness checks. The top candidate was tested using 1,000 label permutations for the interaction LRT, 1,000 bootstrap resamples for coefficient direction and interval stability, a metadata-explanation model for the PC axis, and independent challenge models adding SRS x steroid, SRS x vasopressor, and major comorbidities.

Component	Specification
Outcome	28-day mortality
Treatment contrast	Vasopressin vs noradrenaline
Known endotype adjustment	Published SRS status
Clinical adjustments	Age, APACHE II, sex, steroid exposure
Challenge adjustments	SRS x steroid, SRS x vasopressor, diabetes, immunocompromise, cancer, COPD, renal failure
Discovery status	Retrospective/post-hoc research screen

3. Primary Statistical Results

PC11 was the strongest vasopressin-response candidate among 25 screened transcriptomic PCs. The interaction coefficient was negative, meaning that higher PC11 values were associated with a lower odds ratio for mortality under vasopressin relative to noradrenaline.

Result	Value
Top candidate axis	PC11
Interaction term	Vasopressin x PC11
Beta	-1.779
OR per 1 SD PC11	0.169
95% CI for OR	0.051-0.563
Wald p	0.00377
LRT statistic	10.403
LRT p	0.00126
BH FDR across 25 PCs	0.0315

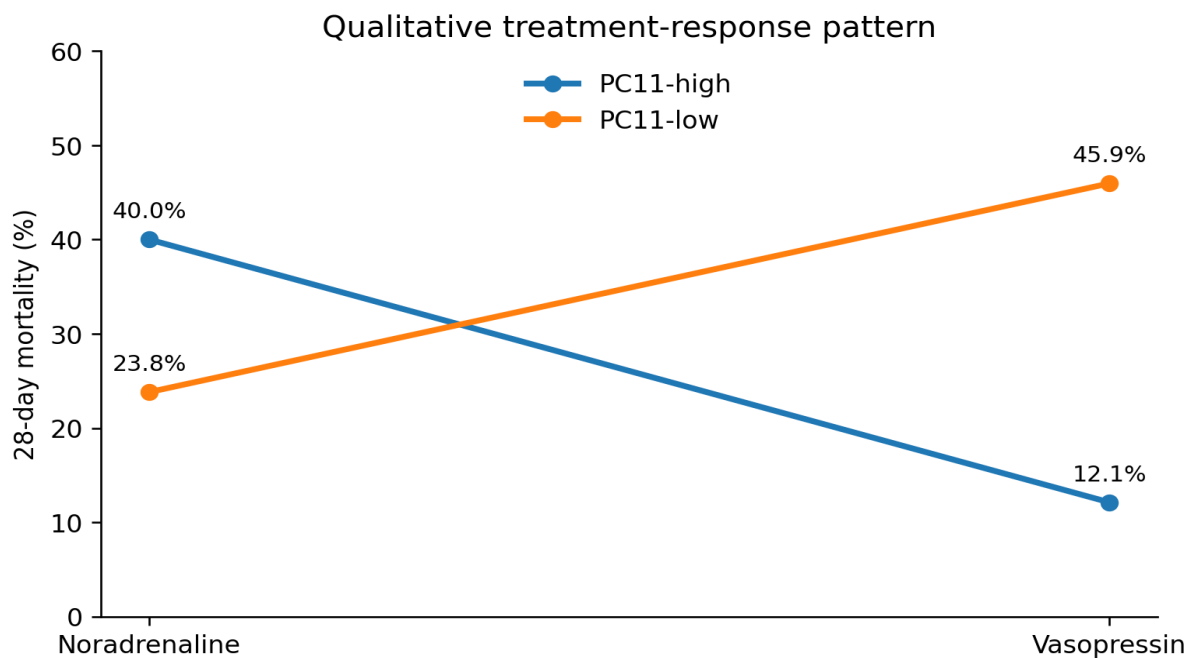


Figure 2. Descriptive interaction pattern. PC11-high shows lower observed mortality with vasopressin, while PC11-low shows higher observed mortality with vasopressin. This plot summarizes observed rates, not a clinical rule.

4. Robustness and Independence Checks

The PC11 x vasopressin interaction remained significant after controlling for known SRS-related interactions and major comorbidities. The point estimate was stable across all challenge models, with ORs ranging from 0.139 to 0.172.

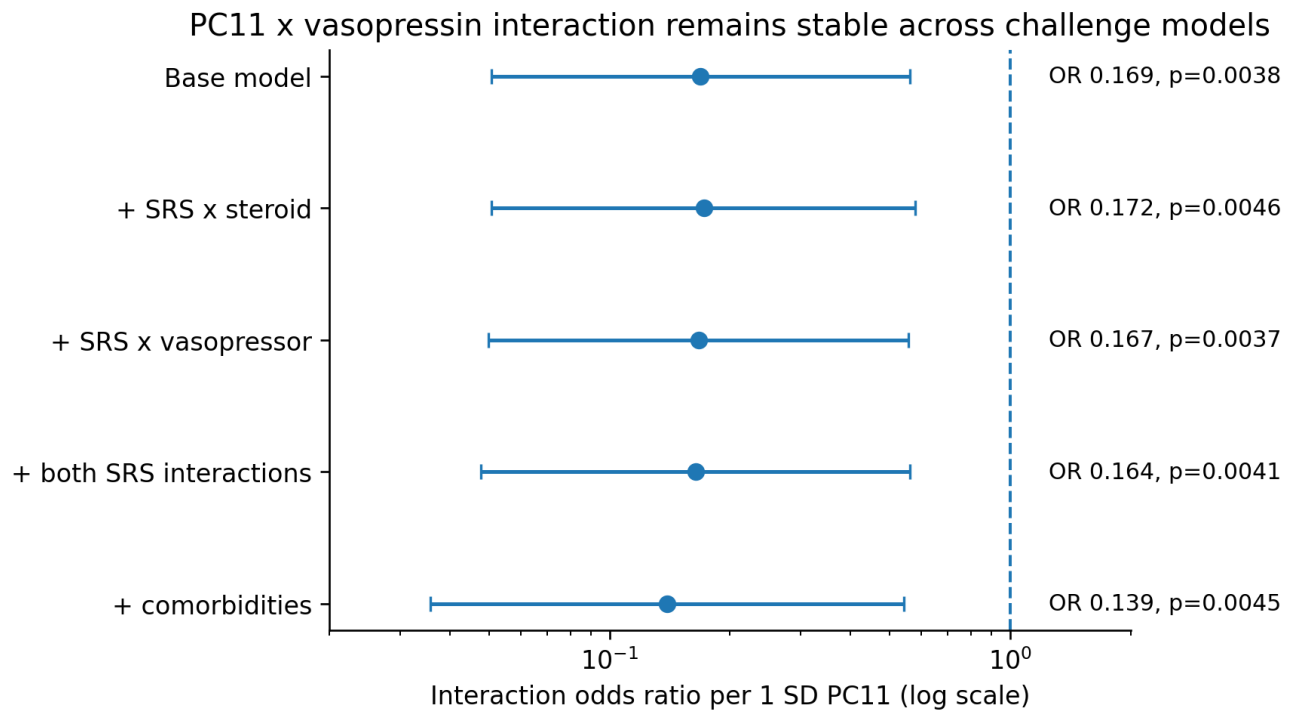


Figure 3. Forest plot of the PC11 x vasopressin interaction across independence challenge models. All models remain below OR=1, supporting a stable interaction direction.

Model	OR	95% CI	p-value
Base model	0.169	0.051-0.562	0.0038
+ SRS x steroid	0.172	0.051-0.581	0.0046
+ SRS x vasopressor	0.167	0.050-0.559	0.0037
+ both SRS interactions	0.164	0.048-0.563	0.0041
+ comorbidities	0.139	0.036-0.544	0.0045

Metadata explanation check. PC11 was not well explained by available metadata: $R^2=0.018$, adjusted $R^2=-0.076$, model $p=0.997$. This does not eliminate all confounding, but it argues against a simple metadata artifact in this audit.

5. Randomization and Bootstrap Evidence

Permutation testing evaluated whether the observed interaction LRT could arise under label-randomized null models. The observed LRT exceeded the null q99 benchmark, with permutation $p=0.0040$.

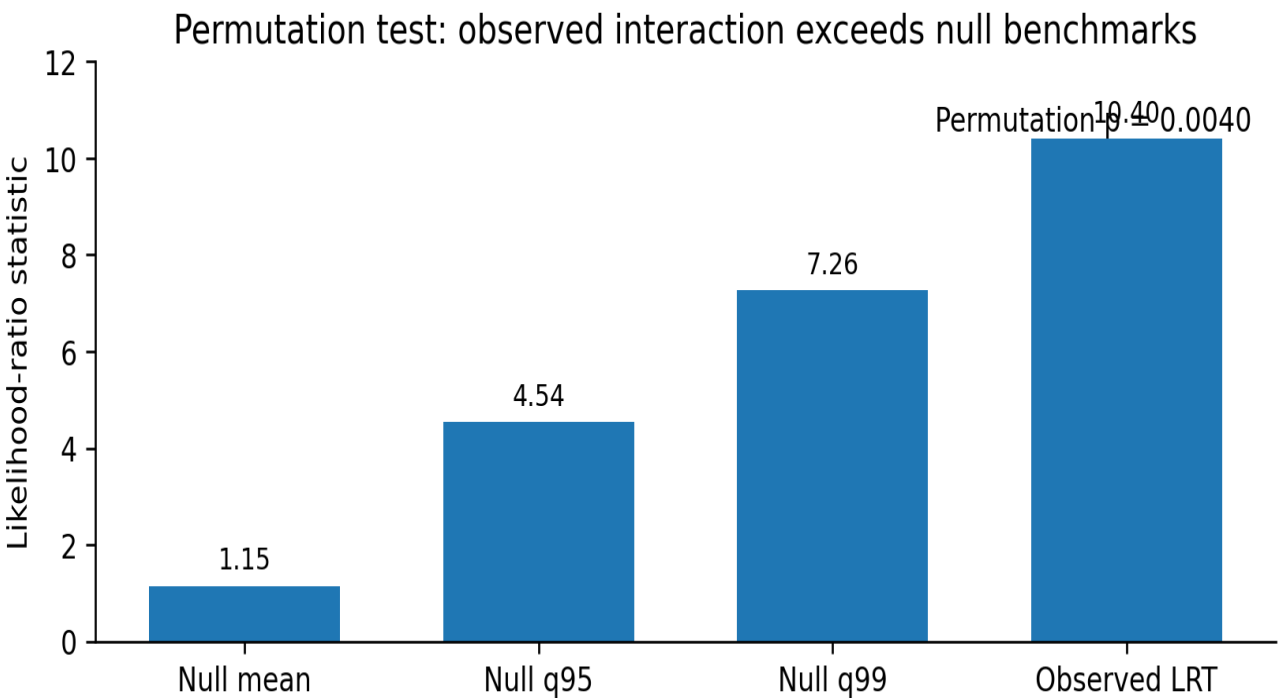


Figure 4. Permutation summary for the top PC11 interaction. The observed LRT is higher than the null q99 benchmark.

5b. Bootstrap Stability Evidence

Bootstrap testing supported coefficient-direction stability. The bootstrap median OR was 0.152, with 5th-95th percentile interval 0.031-0.430. The interaction coefficient was negative in 99.9% of successful bootstrap samples. $p < 0.05$ in 86.1%.

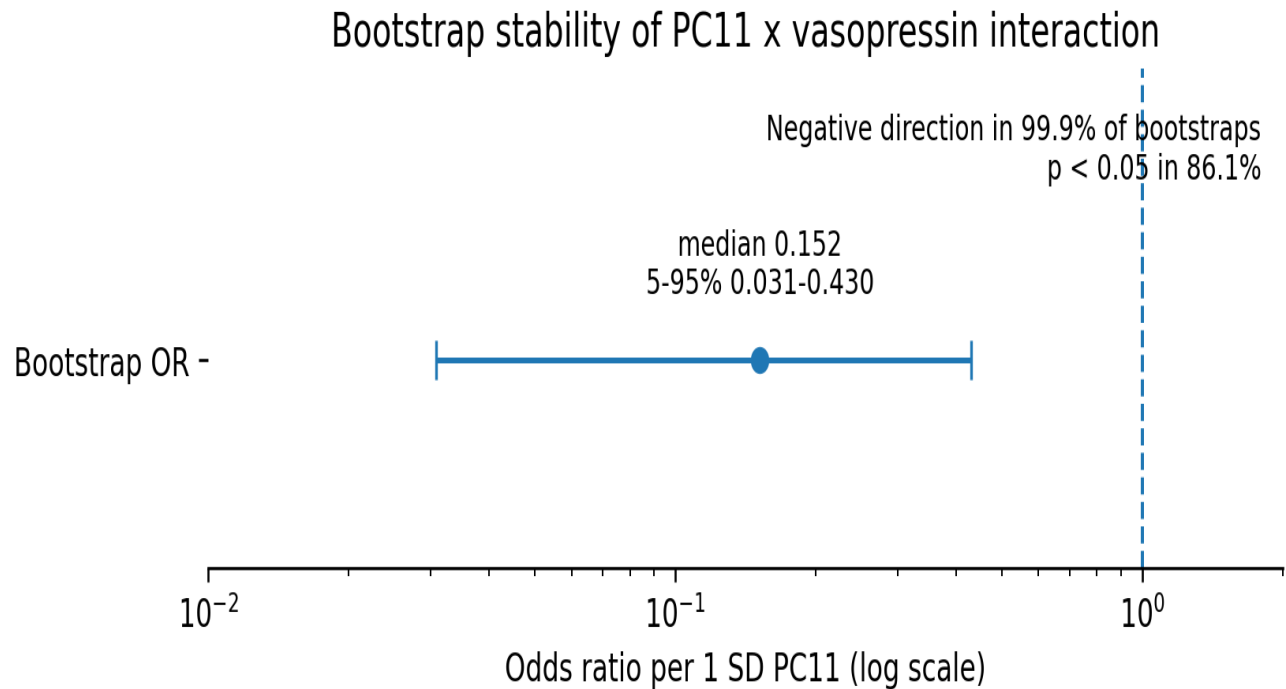


Figure 5. Bootstrap odds-ratio interval for PC11 x vasopressin. The interval remains below OR=1 in the 5th-95th percentile summary.

6. Descriptive Stratification by Steroid Arm

Because VANISH had a factorial design, OmicsTrust also summarized PC11 group, vasopressor, and steroid exposure. These subgroup counts are descriptive and small, but they help identify whether the PC11 vasopressin pattern is driven by an obvious single treatment cell.

PC11 group	Vasopressor	Steroid	Deaths / N	Mortality
PC11-high	Noradrenaline	Placebo	2/11	18.2%
PC11-high	Noradrenaline	Hydrocortisone	8/14	57.1%
PC11-high	Vasopressin	Placebo	2/19	10.5%
PC11-high	Vasopressin	Hydrocortisone	2/14	14.3%
PC11-low	Noradrenaline	Placebo	2/9	22.2%
PC11-low	Noradrenaline	Hydrocortisone	3/12	25.0%
PC11-low	Vasopressin	Placebo	8/19	42.1%
PC11-low	Vasopressin	Hydrocortisone	9/18	50.0%

In PC11-high patients, vasopressin-associated mortality was low in both placebo and hydrocortisone strata. In PC11-low patients, vasopressin-associated mortality was high in both placebo and hydrocortisone strata. However, these cells are small and should be treated as exploratory.

7. Biological Annotation of PC11

The Illumina probe annotation step successfully mapped 500/500 top PC11 probes to the manifest and recovered gene symbols for 446 probes, corresponding to 443 unique symbols. The top annotated genes suggest immune, interferon, myeloid receptor, and stress-response content rather than a purely anonymous PCA artifact.

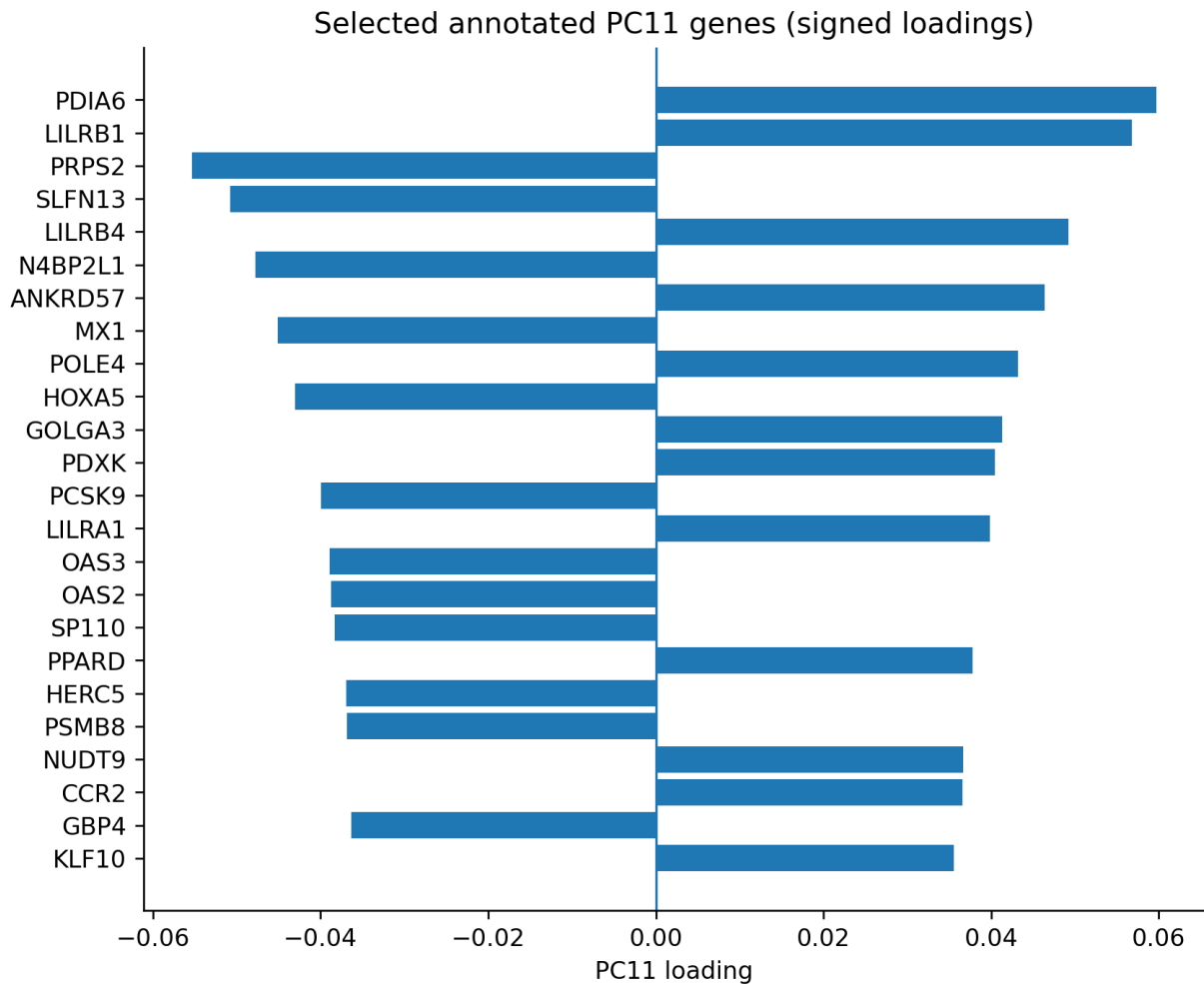


Figure 6. Selected signed PC11 loadings from the annotated top-probe list. Positive and negative signs define opposite ends of the PC11 axis and should be interpreted as axis direction, not as gene-level causal effects.

Signal family	Examples in top annotated probes	Interpretation
Interferon / antiviral innate immunity	MX1, OAS2, OAS3, HERC5, GBP4, SLFN13	Suggests an IFN/innate immune component on one side of PC11.
Antigen processing / immune proteasome	PDIA6	Suggests immune proteasome and antigen-processing biology.
Myeloid immune receptors / trafficking	LILRB1, LILRB4, LILRA1, CCR2	Suggests monocyte/myeloid regulatory biology.
Stress / metabolic / regulatory genes	PDIA6, PPARD, KLF10, PDXK	May indicate stress-response or cell-state contributions.

8. Interpretation for a Clinician or Researcher

What the analysis supports: PC11 is a high-priority candidate transcriptomic treatment-response axis in the VANISH dataset. It appears to identify a subgroup pattern in which vasopressin is associated with lower mortality in PC11-high patients and higher mortality in PC11-low patients, after adjustment for SRS and clinical covariates.

What the analysis does not prove: This report does not prove that PC11 should guide vasopressor therapy in patients. It does not establish causality, clinical utility, or an approved diagnostic threshold. The analysis is retrospective and requires external validation.

Recommended validation plan:

Step	Purpose
Lock the PC11 score	Convert the data-dependent PCA axis into a fixed gene-weighted score that can be applied to new patients.
External cohort validation	Test the locked score in independent septic shock transcriptomic cohorts with vasopressor and mortality data.
Pathway enrichment	Run formal enrichment for positive and negative PC11 gene sets separately.
Cell-composition challenge	Test whether PC11 is explained by leukocyte composition or sample handling.
Prospective utility study	Evaluate whether PC11 stratification improves treatment selection, trial design, or subgroup identification.

Suggested wording: "A de novo OmicsTrust screen of VANISH identified PC11 as a candidate transcriptomic vasopressin-response axis. The interaction remained significant after SRS-interaction and comorbidity challenges and showed permutation and bootstrap support. This is a research-stage finding requiring external validation."

9. Limitations and Safety Statement

Limitation	Implication
Retrospective post-hoc analysis	The result should be treated as hypothesis-generating, not practice-changing.
Moderate sample size (n=116)	Confidence intervals and subgroup rates may be unstable.
PCA axis is cohort-dependent	A locked gene score is required before use in new data.
Residual confounding possible	Metadata and model checks reduce but do not eliminate confounding risk.
Biology requires formal enrichment	Top genes are interpretable but do not prove mechanism.
No clinical threshold established	No patient should be treated differently based only on this report.

Clinical safety statement: This document is for research discussion only. It is not a diagnostic device, not a clinical decision-support tool for bedside vasopressor selection, and not a treatment recommendation. Any clinical use would require locked scoring, independent validation, prospective evaluation, and appropriate regulatory/ethics review.

10. References and Source Notes

#	Reference
1	Gordon AC, Mason AJ, Thirunavukkarasu N, et al. Effect of Early Vasopressin vs Norepinephrine on Kidney Failure in Patients With Septic Shock: The V
2	Antcliffe DB, Burnham KL, Al-Beidh F, et al. Transcriptomic Signatures in Sepsis and a Differential Response to Steroids: From the VANISH Randomized
3	E-MTAB-7581 dataset entry: SRS endotype assignment of septic shock patients from the VANISH randomized trial.
4	OmicsTrust user-provided analysis outputs: vanish_denovo_vasopressin_endotype_discovery.json; vanish_pc11_independence_challenge.json; vanish_

Appendix: exact primary model and key output values

Field	Value
Model	death ~ vasopressor * PC + steroid + SRS + age + APACHE + sex
Top interaction term	C(vasopressor)[T.Vasopressin]:PC11
Observed LRT	10.4029
Reduced vs full model log likelihood	-66.0498 vs -60.8484
Permutation null mean / q95 / q99	1.149 / 4.542 / 7.261
Bootstrap OR median / q05 / q95	0.152 / 0.031 / 0.430
Metadata model R2 / adj R2 / p	0.0179 / -0.0756 / 0.9966